Aayaann Kausar

AI/ML Researcher | LLM Post-Training & Multi-Agent Systems
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Summary

AI researcher, IIT Kharagpur, specializing in LLM post-training: catastrophic forgetting, QLoRA, PEFT and efficient Transformers
Engineered LangGraph/MCP agents with hybrid RAG; formally verified a forgetting bound in Lean 4 and quantized models
Ranked 9th of 5,000+ competitors in a global HFT forecasting challenge; built a real-time speech translator across 7 languages

Education

Indian Institute of Technology (IIT) Kharagpur — Kharagpur, India
Bachelor of Technology | ***Minor in Artificial Intelligence*** — *2024 – Present*

Research Experience

Summer Research Intern | IIIT Bangalore — May 2025 – Jun 2025

Optimized DistilBERT intrusion detection on CICIDS2018 via INT8 quantization, latency profiling and MHA extraction for edge accelerators

- Applied dynamic **INT8 post-training quantization** to DistilBERT linear layers, benchmarked against FP32 on CICIDS2018
- Improved accuracy from **91%** to **96%** across **500 samples** and achieved a **2.41x CPU speedup** over 100 timing runs
- Extracted **MHA** tensors via PyTorch hooks into **ONNX** and INT8 C headers to support a custom Vitis HLS **FPGA** accelerator

Independent Research | Subspace Feature Preservation | Code | Paper — Feb 2026 – Oct 2026

Investigated PCA-ranked activation drift in Qwen2.5-1.5B as a predictor of catastrophic forgetting and a feature-preservation regularizer

- Identified 32 ablation-ranked PCA directions in **Qwen2.5-1.5B** whose drift explains forgetting ($R^2 = 0.933$) versus random axes ($R^2 = 0.850$)
- Proposed rank-32 **SFP regularization** (0.754 retention, 0.650 plasticity), attaining a **0.699 harmonic score** versus 0.631 for LwF
- Formally verified a conditional forgetting bound in **Lean 4** and validated drift predictors across Qwen models of 135M and 1.5B parameters

Scholastic Achievements

JEE Main 2024 — Top **0.29%** of 1.4M+ candidates in India's highly competitive national engineering entrance examination
JEE Advanced 2024 — Top **3.2%** of 250K+ aspirants in one of the world's most competitive undergraduate entrance examinations
WBJEE 2024 — Top **0.20%** among 113K+ candidates in West Bengal's highly competitive state engineering entrance examination

Projects

Real-Time Voice Translator | Multilingual Speech AI | Demo | Code — Sep 2026 – Oct 2026

Engineered a live 7-language speech translator with Next.js, FastAPI WebSockets, Whisper, GPT-OSS 20B and ElevenLabs TTS

- Built a streaming ASR-MT-TTS pipeline for bidirectional **7-language** translation with Whisper, GPT-OSS 20B and ElevenLabs Flash
- Implemented 20 ms RMS voice activity detection with 600 ms endpointing, polyphase resampling and stale-response filtering
- Deployed a **Next.js 16 AudioWorklet** client on Vercel and Render with bilingual captions, p50/p95 latency telemetry and 27 backend tests

Qwen2.5 | QLoRA Fine-Tuning Study | Code — Aug 2026 – Sep 2026

Evaluated 4-bit QLoRA adaptation of Qwen2.5-1.5B via controlled held-out experiments, task-complexity stratification and error analysis

- Fine-tuned **Qwen2.5-1.5B** on 1,286 ML systems tasks using **4-bit NF4 QLoRA** with rank-16 adapters, TRL, PEFT and 8-bit AdamW
- Achieved a **30.42% ROUGE-L gain** on 161 held-out tasks while raising BERTScore F1 from 0.8187 to 0.8497 post-adaptation
- Stratified results across 48 topics and difficulty tiers: **34.84% higher Staff ROUGE-L**, with 14.2% of held-out tasks regressing

NexusAgent | Enterprise Decision Intelligence Platform | Code — Aug 2026 – Sep 2026

Engineered a LangGraph multi-agent decision platform integrating hybrid RAG, secured MCP NL2SQL and grounded answer validation

- Optimized hybrid retrieval with **Milvus**, BM25, RRF and BGE reranking, lifting Hit@1 from **85.62% to 96.25%** and MRR@5 to **98.12%**
- Hardened MCP NL2SQL with **SQLGlot** AST validation, read-only scopes and query timeouts, passing all **28 security boundary tests**
- Orchestrated 3 LangGraph agent routes with an evidence-review stage, validated by **1,941 unit** and 323 offline integration tests

Competitions

HFT Market State Forecasting Challenge | Wunder Fund — Sep 2025 – Nov 2025

Forecasted high-frequency market state vectors over 1,000-step streaming sequences using temporal representation learning and low-latency inference

- Ranked **9th of 5,000+ competitors** in the global HFT forecasting challenge, achieving final R^2 of 0.3920 (public score 0.3880)
- Pretrained a **4-layer, 8-head Transformer** using masked autoencoding, temporal positional encodings and robust median-IQR scaling
- Accelerated single-core CPU inference via **torch.compile**, Numba kernels and preallocated buffers for low-latency streaming forecasts

Skills & Expertise

Languages & Core: Python, C++, CUDA, Triton, SQL, Bash, Linux, Git, NumPy, Pandas, JavaScript, TypeScript, WebSockets
AI/ML & Research: PyTorch, Hugging Face, Transformers, PEFT, QLoRA, TRL, Lean 4, ROUGE-L, BERTScore F1, Whisper
Agents & Systems: LangGraph, LangChain, LangSmith, MCP, Hybrid RAG, FastAPI, Docker, Postgres, Next.js 16, ElevenLabs

Coursework

Mathematics: Advanced Calculus | Probability and Statistics | Linear Algebra | Numerical Analysis

CS: Data Structures and Algorithms **AI:** Deep Learning Foundations | Reinforcement Learning | Foundations of LLMs

MOOCs: Language Modeling from Scratch (**Stanford CS336**)